Type MS · Explain how a mutation changes a protein and phenotype

Answer 1 of 10 · Question 7 part a · 3 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

Number				
7	(a)	Any three from: • changes the DNA/base/codon sequence; • changes the order of the amino acids/different amino acid coded for; • changes the function/shape of the protein/different protein; • affects a characteristic or function/genotype in the individual;		3
	(b) (i)	Two from: • dialysis has to be completed several times a week/transplant permanent; • increased life expectancy; • less restrictions on diet; • better quality of life/freedom to move around/normal lifestyle;		2
	(ii)	100 x 3233÷4733; = 68.3/68		2
	(iii)	Two from: • people have two kidneys; • two recipients from one donor; • living donation/can live with just one;		2
	(c) (i)	0.5:1;	Accept 1:2	1
	(ii)	individual B individual A D: Dd, Dd d: dd, dd • correct genotype individual A; • correct genotype individual B; • 50% offspring heterozygous/correct genotype and affected; • 50% homozygous recessive/correct genotype and not affected;		4

Total 14 marks

Type MS · Explain how a mutation changes a protein and phenotype

Answer 1 of 10 · Question 7 part a · 3 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

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Type MS · Explain how a mutation changes a protein and phenotype

Answer 2 of 10 · Question 8 part c · 4 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

number				
8 (a)	(i)	absorbs uv light; reduces/prevents risk of skin cancer;	Allow protects against (UV) radiation	2
	(ii)	eye/ retina/ choroid/ red blood cell/ haemoglobin ;		1
(b)	(i)	mother x father Hh hh; gametes H h h; fertilisation Hh hh; phenotype white forelock normal;	Reject diagrams that show sex linkage	4
	(ii)	chance of producing a boy is 0.5/½; chance of producing offspring with condition 0.5/½; chance of producing boy with condition is ½ × ½ = ¼;	Allow 50% or equivalent anywhere in calculation Full marks for correct final answer	3
(c)		Any 4 from different (genetic) code produced; different order of amino acids; codes for different proteins/enzymes; enzyme substrate complex not formed; causes change in pigment;	Allow change in base sequence/genetic code Allow any reference to change in amino acids	4

Total 14 marks

Type MS · Explain how a mutation changes a protein and phenotype

Answer 2 of 10 · Question 8 part c · 4 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

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Type MS · Explain how a mutation changes a protein and phenotype

Answer 3 of 10 · Question 5 part a · 3 marks

2022 JAN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

Question number				
5	(a)	• change in base sequence; • causes change in protein synthesis; • causes change in phenotype;		3
	(b) (i)	• axes labelled with units; • suitable scale; • correct plots; (3 x 2) • bars of equal width;		5
	(ii)	• 43 000 × 0.0015; • = 64.5; • 65;		3
	(c)	• extra chromosome at pair 21; • diploid number 47; • female because 2 X chromosomes; • at pair 23;		4

Total : 15 marks

Type MS · Explain how a mutation changes a protein and phenotype

Answer 3 of 10 · Question 5 part a · 3 marks

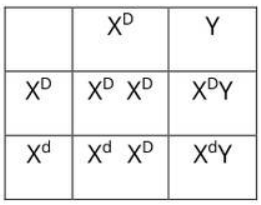
2022 JAN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how a mutation changes a protein and phenotype

Answer 4 of 10 · Question 8 part a · 4 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

8 (a)	any 4 from • change in base sequence produces different mRNA; • altered codon (or codons); • so different amino acids coded; • incorrect protein made; • could produce a stop codon so no protein made/faulty protein made;		4
(b) (i)	• faulty allele present on one chromosome only; • effects overridden by a dominant allele;		2
(ii)	• one male in generation 1 has DMD; • (faulty) allele must be on X chromosome; • male is XY/does not carry dominant allele on Y chromosome; • carriers present;		3
(iii)	• correct parental genotypes; • correct genotypes for generation 1; 		2
			Total 11

Type MS · Explain how a mutation changes a protein and phenotype

Answer 4 of 10 · Question 8 part a · 4 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how a mutation changes a protein and phenotype

Answer 5 of 10 · Question 6 part a.i · 4 marks

2023 JUN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

Question number				
6	(a) (i)	• change in genetic code/base sequence/nucleotide sequence/different codon; • of DNA/gene; • different amino acid coded for/different sequence of amino acids; • different (shaped) protein produced;	Allow named mutation e.g. deletion	4
	(ii)	Any five from • shape of red blood cell distorted; • different form of haemoglobin; • reduced surface area; • less oxygen carried; • less respiration; • less energy/ATP;	Allow red blood cells are sickle-shaped Allow capillaries blocked/allow less haemoglobin	5
	(b)	• woman is Hh; • man is either Hh or hh/heterozygous or homozygous recessive; • one h/recessive allele comes from man; • child is hh/homozygous recessive;		4

Total 13

Type MS · Explain how a mutation changes a protein and phenotype

Answer 5 of 10 · Question 6 part a.i · 4 marks

2023 JUN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how a mutation changes a protein and phenotype

Answer 6 of 10 · Question 5 part b.i · 3 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

Number				
5	(a)	4^3; = 64;	Full marks for correct answer	2
	(b) (i)	• change in order/sequence; • of bases; • different protein may be formed;		3
	(ii)	uv light/λ rays/ionising radiation/X-rays;	accept any suitable example	1
	(iii)	• can't catalyse reaction/slows reaction/activity; • because can't form E-S complexes/substrate can't bind with active site/active site not complementary;		2
	(iv)	doesn't affect working of enzyme/mutation does not change amino acid/doesn't affect structure/shape of enzyme/different coding for same amino acid;		1

Total 9 marks

Type MS · Explain how a mutation changes a protein and phenotype

Answer 6 of 10 · Question 5 part b.i · 3 marks

2023 NOV · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how a mutation changes a protein and phenotype

Answer 7 of 10 · Question 5 part b.iv · 1 mark

2023 NOV · Paper 01 · extract 1 of 2 · find part b.iv in this mark scheme extract

Number				
5	(a)	4^3; = 64;	Full marks for correct answer	2
	(b) (i)	• change in order/sequence; • of bases; • different protein may be formed;		3
	(ii)	uv light/λ rays/ionising radiation/X-rays;	accept any suitable example	1
	(iii)	• can't catalyse reaction/slows reaction/activity; • because can't form E-S complexes/substrate can't bind with active site/active site not complementary;		2
	(iv)	doesn't affect working of enzyme/mutation does not change amino acid/doesn't affect structure/shape of enzyme/different coding for same amino acid;		1

Total 9 marks

Type MS · Explain how a mutation changes a protein and phenotype

Answer 7 of 10 · Question 5 part b.iv · 1 mark

2023 NOV · Paper 01 · extract 2 of 2 · find part b.iv in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how a mutation changes a protein and phenotype

Answer 8 of 10 · Question 2 part b · 3 marks

2024 JUN · Paper 02 · find part b in this mark scheme extract

Number				
2 (a)	(i)	• nucleus; • cytoplasm/RER/ribosomes;	Allow mitochondrion as an alternative mp	(2)
	(ii)	Any two from: • RNA is single stranded/DNA double stranded; • RNA contains uracil/DNA contains thymine ; • RNA contains ribose /DNA contains deoxyribose;	Allow DNA is double helix Do not allow thiamine	(2)
	(iii)	nucléotide ;	Allow mononucleotides	(1)
(b)		Any three from • incorrect base/ nucleotide/codon sequence; • incorrect amino acid sequence/amino acid; • faulty/different/incorrect protein made; • reference to protein being a different shape/has a different function;	Allow named mutation/substitution/ deletion/addition/ insertion/eq	(3)

Total for Question 2 = 8 marks

Type MS · Explain how a mutation changes a protein and phenotype

Answer 9 of 10 · Question 5 part a.ii · 3 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

5 (a)	(i)	• change in nucleotide/base/codon sequence; • of DNA;		2
	(ii)	• new/different (sequence of) amino acids; • different protein / enzyme / different polypeptide • different phenotype/enzyme does not bind to substrate;		3
(b)	(i)	• allele for haemophilia on X chromosome/ not on Y chromosomes; • hairy ears could only be males / haemophilia could occur in females (and males);		2
	(ii)	• 1 = XX; • 2 = XY^H;		2
	(iii)	• chance of being a girl = 50%/ this can be indicated in a punnet square; • girl will have XX genotype / girl indicated as XX from punnet square/ girls have two X; • no Y chromosome so no chance of hairy ears;		3

Total 12 marks

Type MS · Explain how a mutation changes a protein and phenotype

Answer 9 of 10 · Question 5 part a.ii · 3 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how a mutation changes a protein and phenotype

Answer 10 of 10 · Question 7 part a.i · 2 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

7 (a) (i)	Any two from the following • causes cancer; • gene mutation; • incorrect protein made/shape of protein incorrect;		2
(ii)	• changing the way free radicals behave; • to allow normal flow of substances/named substance in and out of cells;		2
(iii)	• 40g = 60% of daily requirement; • $\frac{40 \times 100}{60}$; • = 67g;	Allow 66.7g	3
(b)	• add fresh orange juice to a test tube/named suitable container; • add DCPIP drop by drop until it no longer decolourises; • record the volume of DCPIP added; • repeat with the same volume of boiled orange juice; • compare volumes;		5
			Total 12

Type MS · Explain how a mutation changes a protein and phenotype

Answer 10 of 10 · Question 7 part a.i · 2 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks